NILESH KUMAR

Senior Software Engineer

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Targeting a Software Development Engineer role in a growth-driven technology organization, with a focus on building scalable, resilient, and high-performance systems using Java, Spring Boot, React and cloud-native architectures.

PROFILE SUMMARY

- **Senior Software Engineer with 5 years of experience in building end-to-end healthcare imaging systems using Java, Spring Boot, React, and AWS.**
- Experienced in developing **frontend modules** (React, Angular) aligned with radiology workflows such as patient registration, local data visualization, and image distribution.
- Engineered distributed systems **processing over 3M+ daily transactions** with robust fault tolerance and dynamic scaling.
- **Reduced workflow latency by 35%+** via advanced parallelization, **Kafka**-based messaging, and optimized **REST APIs**.
- Drove migration from monolith to Microservices, **cutting release cycles by 40%** and boosting service reliability.
- **Lowered infrastructure costs by 30%** through **AWS Lambda** integration, **Docker** containerization, and **resource throttling**.
- Strengthened CI/CD practices using Jenkins, **achieving 99% uptime** and improving deployment success rates.
- Proactively **expanded test coverage by 23%** and enforced clean code reviews, reducing bugs and technical debt.
- Adept at **integrating complex backend logic with interactive UI components**, improving usability and system responsiveness

CORE COMPETENCIES

- Backend Architecture Design
- Fault-Tolerant System Design
- Clean Code & Code Review Practices
- Innovation & Prototyping
- Performance Tuning & Optimization
- Continuous Integration & Deployment
- Test Automation & Coverage Expansion
- AI Integration in Backend Systems
- Scalable Microservices Development
- Cloud Cost Optimization
- Technical Mentorship
- Full Stack Application Development

SOFT SKILLS



TECHNICAL SKILLS

Languages: Java, SQL
Frameworks: Spring Boot, Hibernate, Microservices
Cloud & DevOps: AWS (Lambda, S3, CloudWatch), Docker, Jenkins, CI/CD
Data & Messaging: Kafka, MySQL, RESTful APIs
Testing: JUnit, Mockito
Frontend & UI: React JS(Hooks, Context API, Axios), Angular, Thymeleaf

EDUCATION

2020: B.E. Electronics and Communication, Siddaganga Institute of Technology, Tumkur | CGPA: 7.76

WORK EXPERIENCE

Siemens Healthineers, Bangalore

Senior Software Engineer — **Sept 2024 - Present**

- Optimizing Java/Spring Boot workflow to reduce startup time by 2 seconds per request, saving 70+ hours/year.
- Enhanced frontend performance through lazy loading and API pagination, reducing UI load time by 50%, improving technician productivity during high-volume scans.
- Implementing memory throttling logic, cutting GPU resource crashes by 45% during high-load imaging workflow.
- Automating test coverage expansion by 23% using JUnit/Mockito, minimizing post-deployment errors.
- Developed React-based Distribution Screen to display patient-wise image thumbnails with on-click preview functionality, enabling radiologists to quickly verify images before PACS transfer.Leading PR reviews and clean code enforcement, reducing tech debt by 30%.

Software Engineer — **Jul 2022 – Aug 2024**

- Designed Kafka-based retry logic, minimizing manual image re-uploads by 60%.
- Migrated monolith services to Spring Boot Microservices, enhancing modularity and reducing release time by 40%.
- Built React-based Local Data Screen that dynamically fetched and displayed acquired image lists, giving technologists real-time visibility into exam completeness.
- Added Angular-based configuration module to modify scanning parameters, reducing dependency on manual XML configurations and enhancing usability for field engineers.
- Built CI/CD pipelines in Jenkins, ensuring 99% deployment reliability.
- Refined SQL performance, improving reporting by 15% and prioritizing real-time job queues by urgency.

Associate Software Engineer **Sept 2020 – Jun 2022**

- Automated post-upgrade validation tools, reducing downtime by 40% & saving $12K/year.
- Developed centralized log cleanup utilities, reducing system diagnosis time by 25%.
- Implemented React-based Patient Registration Screen to capture and validate patient data before examination, improving registration accuracy by 30%.
- Collaborated with UI/UX team to make forms more intuitive and accessible, reducing technician data entry errors and speeding up exam initiation.
- Prototyped ML-based image screening tools, automating 20% of manual validation efforts.
- Delivered RESTful APIs for imaging device configuration, streamlining setup workflow.

INDEPENDENT PROJECTS

Production-Ready Patient Management System

- Designed and deployed a microservices-based backend for managing patient records, appointments, and medical reports. Emphasized modular architecture, fault-tolerant service communication, and CI/CD readiness.
- Tech Stack: Spring Boot Microservices, Spring Data, Spring Cloud (Eureka, Gateway, Circuit Breaker), Kafka, PostgreSQL, Docker

Audio Transcriber (Spring AI & React)

- Developed a full-stack audio transcription tool allowing users to upload audio files, transcribe them via AI APIs, and view/store the results through a user-friendly interface.
- Tech Stack: Spring Boot, Spring Web, Spring Data JPA, AI Transcription APIs (OpenAI/Google), React JS, REST APIs, MySQL/PostgreSQL, Docker

AI-Powered Smart Email Assistant

- Engineered a contextual email reply generator using AI for natural language processing. Integrated backend logic with Google Gemini API and Chrome extension to embed AI-generated replies within Gmail.
- Tech Stack: Java, Spring Boot, Spring AI, Google Gemini API, React JS, REST APIs, Chrome Extension, JSON Parsing

Real-Time Chat Application

- Built a live chat platform with real-time message delivery using WebSockets. Backend manages user sessions and message broadcasting, while Thymeleaf supports dynamic frontend rendering.
- Tech Stack: Java, Spring Boot, Spring WebSocket, Thymeleaf, Maven

PERSONAL DETAILS

Date of Birth: 17th July 1997
Languages Known: English, Hindi, Kannada
Address: Bangalore, Karnataka, 560102